Quantum theory of radiation

Hamiltonian of charged particle in EM field

$$\hat{H} = \frac{1}{2m}(\hat{p} - q\hat{A})^2 + q\varphi - \vec{\mu}\cdot\vec{B} =$$

$$= \frac{\hat{p}^2}{2m} - \frac{q}{2m}(\hat{p}\cdot\hat{A} + \hat{A}\cdot\hat{p}) + \frac{q^2}{2m}\hat{A}^2 + q\varphi - \vec{\mu}\cdot\vec{B}$$

Coulomb gauge: $\varphi = 0$

$$\hat{A}(\vec{r},t) = \sum_{\vec{k}\sigma}\sqrt{\frac{\hbar}{2\varepsilon_0 V\omega_k}}\left\{\hat{a}_{k\sigma}\, e^{-i\omega_k t + i\vec{k}\cdot\vec{r}}\,\hat{e}_{k\sigma} + \hat{a}^+_{k\sigma}\, e^{i\omega_k t - i\vec{k}\cdot\vec{r}}\,\hat{e}^*_{k\sigma}\right\} =$$

$$= \sum_{\vec{k}\sigma}\left(\hat{a}_{k\sigma}\,\vec{A}_{k\sigma}(\vec{r})\,e^{-i\omega_k t} + \hat{a}^+_{k\sigma}\,\vec{A}^*_{k\sigma}(\vec{r})\,e^{i\omega_k t}\right).$$

$$\vec{A}_{k\sigma}(\vec{r}) = \hat{e}_{k\sigma}\sqrt{\frac{\hbar}{2\varepsilon_0 V\omega_k}}\; e^{i\vec{k}\cdot\vec{r}}$$

$$\hat{p} = -i\hbar\vec{\nabla} \Rightarrow \hat{p}\cdot\hat{A} = -i\hbar\frac{\partial}{\partial\vec{r}}\sum_{\vec{k}\sigma}\left(\hat{a}_{k\sigma}\,\vec{A}_{k\sigma}(\vec{r})\,e^{-i\omega_k t} + \hat{a}^+_{k\sigma}\,\vec{A}^*_{k\sigma}(\vec{r})\,e^{i\omega_k t}\right)$$

but $\frac{\partial}{\partial\vec{r}}\,\hat{e}_{k\sigma}\,e^{i\vec{k}\cdot\vec{r}} = \hat{e}_{k\sigma}\cdot(i\vec{k}) = 0$

$\Rightarrow$ $\hat{p}$ and $\hat{A}$ commute with each other.

$$\hat{H} = \hat{H}_0 + \hat{V}(t), \quad \hat{H}_0 = \frac{\hat{p}^2}{2m} + U(r)$$

↳ coulomb potential of nuclei, etc.

time-dependent perturbation:

$$\hat{V}(t) = \underbrace{-\frac{q}{m}\hat{A}\cdot\hat{p}}_{\text{the main 1-photon process}} + \underbrace{\frac{q^2}{2m}\hat{A}^2}_{\text{2-photon process}} \underbrace{- \vec{\mu}\cdot(\vec{\nabla}\times\vec{A})}_{\text{1-photon but usually small (non-relativistic approx.)}}$$

Quick sketch of Fermi's Golden Rule:

$$\hat{H} = \hat{H}_0 + \hat{V}(t), \quad i\hbar \frac{\partial |\psi\rangle}{\partial t} = \hat{H}|\psi\rangle$$

Complete set of states $\hat{H}_0 \psi_n = E_n \psi_n$, $\langle \psi_n | \psi_m \rangle = \delta_{nm}$

let $|\psi(t)\rangle = \sum_n a_n(t) |\psi_n\rangle e^{-iE_n t/\hbar}$

$$\Rightarrow \quad i\hbar \sum_n \left(\frac{\partial a_n(t)}{\partial t} |\psi_n\rangle e^{-iE_n t/\hbar} + a_n(t) |\psi_n\rangle \left(-\frac{iE_n}{\hbar}\right) e^{-\frac{iE_n t}{\hbar}} \right) =$$

$$= \sum_n \left(\hat{H}_0 + \hat{V}(t)\right) a_n(t) |\psi_n\rangle e^{-iE_n t/\hbar}$$

$$\sum_n \left(i\hbar \frac{da_n}{dt} |\psi_n\rangle e^{-iE_n t/\hbar} + E_n |\psi_n\rangle a_n(t) e^{-iE_n t/\hbar} \right) =$$

$$= \sum_n \underbrace{\hat{H}_0 |\psi_n\rangle}_{E_n|\psi_n\rangle} a_n(t) e^{-iE_n t/\hbar} + \hat{V} a_n(t) |\psi_n\rangle e^{-iE_n t/\hbar}$$

Multiply both sides by $\langle \psi_m |$ and use completeness:

$$i\hbar \frac{da_m(t)}{dt} e^{-iE_m t/\hbar} = \sum_n \langle \psi_m | \hat{V}(t) | \psi_n \rangle a_n(t) e^{-iE_n t/\hbar}$$

$$\Rightarrow \quad i\hbar \frac{da_m}{dt} = \sum_n V_{mn}(t)\, a_n(t)\, e^{i\omega_{mn} t}$$

$$\hbar \omega_{mn} = E_m - E_n \; ; \quad V_{mn} \equiv \langle \psi_m | \hat{V} | \psi_n \rangle.$$

matrix element of $n \to m$ transition.

The idea is to solve eq. for $\frac{da_m}{dt}$ perturbatively: (V_{mn} is "small")

system is initially in i-th stationary state $\Rightarrow a_n^{(0)} = \delta_{ni}$

On the RHS: $a_m = a_m^{(0)} + a_m^{(1)} = a_m^{(1)}$

$$i\hbar \frac{d a_m^{(1)}}{dt} = V_{mi}(t)\, e^{i\omega_{mi} t}$$

$$a_m^{(1)}(t) = -\frac{i}{\hbar}\int_0^t dt'\, V_{mi}(t')\, e^{i\omega_{mi} t'}$$

Our perturbation is given by $\hat{V}(t) = -\frac{q}{m}\hat{p}\hat{A}$ and

$$\hat{A} = \sum_{\vec{k}\sigma}\Big(\underbrace{\hat{a}_{k\sigma}\vec{A}_{k\sigma}}_{\hat{V}_{k\sigma}} e^{-i\omega_k t} + \underbrace{\hat{a}^+_{k\sigma}\vec{A}^*_{k\sigma}}_{\hat{V}^+_{k\sigma}} e^{i\omega_k t}\Big)$$

$$a_m^{(1)}(t) = -\frac{i}{\hbar}\sum_{k\sigma}\int_0^t dt'\Big[(\hat{V}_{k\sigma})_{mi}\, e^{i(\omega_{mi}-\omega_k)t'} + (\hat{V}^+_{k\sigma})_{mi}\, e^{i(\omega_{mi}+\omega)t'}$$

$$= -\sum_{k\sigma}\left\{(\hat{V}_{k\sigma})_{mi}\frac{\left(e^{i(\omega_{mi}-\omega_k)t}-1\right)}{\hbar(\omega_{mi}-\omega)} + (\hat{V}^+_{k\sigma})_{mi}\frac{e^{i(\omega_{mi}+\omega_k)t}-1}{\hbar(\omega_{mi}+\omega)}\right\}$$

Transition probability in time t is $|a_{mi}|^2 =$

$$= \sum_{\vec{k}\sigma}\Bigg\{\left|(\hat{V}_{k\sigma})_{mi}\right|^2\frac{\sin^2\frac{(\omega_{mi}-\omega_k)t}{2}}{\hbar^2\left(\frac{\omega_{mi}-\omega}{2}\right)^2} + \left|(\hat{V}^+_{k\sigma})_{mi}\right|^2\frac{\sin^2\frac{(\omega_{mi}+\omega_k)t}{2}}{\hbar^2\left(\frac{\omega_{mi}+\omega}{2}\right)^2}$$

$$+ (\hat{V}_{k\sigma})_{mi}(\hat{V}^+_{k\sigma})^*_{mi}\frac{\left(e^{i(\omega_{mi}-\omega_k)t}-1\right)\left(e^{-i(\omega_{mi}+\omega_k)t}-1\right)}{\hbar^2(\omega_{mi}-\omega_k)(\omega_{mi}+\omega_k)}$$

$$+ (\hat{V}^+_{k\sigma})_{mi}(\hat{V}_{k\sigma})^*_{mi}\frac{\left(e^{i(\omega_{mi}+\omega_k)t}-1\right)\left(e^{-i(\omega_{mi}-\omega_k)t}-1\right)}{\hbar^2(\omega_{mi}-\omega_k)(\omega_{mi}+\omega_k)}\Bigg\}$$

In the first two terms we used

$$\left(e^{i\Omega t}-1\right)\left(e^{-i\Omega t}-1\right) = 2\left(1-\cos\Omega t\right) = 4\sin^2\left(\frac{\Omega t}{2}\right)$$

for $\Omega = \omega_{mi} \pm \omega_k$.

We next observe that

$$\lim_{t\to\infty} \frac{\sin^2\left(\frac{\Omega t}{2}\right)}{t\,(\Omega/2)^2} = \pi\,\delta\left(\frac{\Omega}{2}\right) = 2\pi\,\delta(\Omega)$$

Indeed, for $\Omega \neq 0$ $\lim_{t\to\infty} \frac{\sin^2(\Omega t/2)}{t\,(\Omega/2)^2} = 0$

while for $\Omega = 0$ $\lim_{t\to\infty} \frac{\sin^2(\Omega t/2)}{t\,(\Omega/2)^2} = \lim_{t\to\infty} t = \infty.$

$\Rightarrow$ $\frac{\sin^2(\Omega t/2)}{t\,(\Omega/2)^2} = c\,\delta(\Omega)$ and we can fix c by integrating both sides over $\int_{-\infty}^{\infty} d\Omega$.

$$\Rightarrow c = 2\pi. \quad \left(\text{Using } \int_{-\infty}^{\infty} dx\, \frac{\sin^2(x)}{x^2} = \pi.\right)$$

Therefore $|a_{mi}|^2 = \sum_{k\sigma} \Big\{ \left|(\hat{v}_{k\sigma})_{mi}\right|^2 \frac{2\pi}{\hbar^2}\,\delta(\omega_{mi} - \omega_k)\,t$

$$+ \frac{2\pi}{\hbar^2} \left|(\hat{v}^{+}_{k\sigma})_{mi}\right|^2 \delta(\omega_{mi} + \omega_k)\,t + \left(\begin{array}{l}\text{terms which } \underline{\text{do not}} \\ \text{grow with time } t\end{array}\right)\Big\}$$

Therefore for large times t we can define rate of ~~the~~ transition between initial state i and final state $f = m$:

Fermi's Golden Rule

$$dW_{fi} = \frac{|a_{mi}|^2}{t} \to \frac{2\pi}{\hbar} \sum_{\vec{k}\sigma} \Big\{ \left|(\hat{v}_{k\sigma})_{fi}\right|^2 \delta(E_f - E_i - \hbar\omega_k)$$

$$+ \left|(\hat{v}^{+}_{k\sigma})_{fi}\right|^2 \delta(E_f - E_i + \hbar\omega_k) \Big\}$$

Observe that $\left|(\hat{v})_{fi}\right|^2 = v^{*}_{fi}\, v_{fi} = \hat{v}^{+}_{if} \left(\hat{v}^{+}_{if}\right)^{*} = \left|\hat{v}^{+}_{if}\right|^2$

so that $|\hat{v}_{fi}|^2 = |\hat{v}^{+}_{fi}|^2$ (detailed balance).

Term with $\delta(E_f - E_i - \hbar\omega_k)$ describes final state $E_f = E_i + \hbar\omega_k$ of higher than initial energy $\Rightarrow$ this is <u>absorption</u>.

Similarly, for $\delta(E_f - E_i + \hbar\omega_k)$ we find $E_f = E_i - \hbar\omega_k$ $\Rightarrow$ <u>emission</u>.

For the emission $\begin{cases} |i\rangle = \psi_i \, \phi_N & \text{electron in } \psi_i, \text{ N photons} \\ |f\rangle = \psi_f \, \phi_{N+1} & \text{— — } \psi_f, \text{ N+1 photons} \end{cases}$

$$\left(\hat{V}^{+}_{k\sigma}\right)_{fi} = \langle \phi_{N+1} | \langle \psi_f | \hat{a}^{+}_{k\sigma} \vec{A}^{*}_{k\sigma}(r) \left(\frac{-q\hat{p}}{m}\right) | \psi_i \rangle | \phi_N \rangle$$

$$= \underbrace{\langle \phi_{N+1} | \hat{a}^{+}_{k\sigma} | \phi_N \rangle}_{\sqrt{N_{k\sigma}+1}} \langle \psi_f | \vec{A}^{*}_{k\sigma}(r) \left(\frac{-q\hat{p}}{m}\right) | \psi_i \rangle$$

Noting that $\sum_{\vec{k}} \ldots = \frac{V}{(2\pi)^3} \int d^3\vec{k} \ldots$

because $\vec{k} = \left(\frac{2\pi}{L_x} n_x, \frac{2\pi}{L_y} n_y, \frac{2\pi}{L_z} n_z\right)$ and $V = L_x L_y L_z$

$$W^{emi}_{fi} = \frac{2\pi}{\hbar} \frac{V}{(2\pi)^3} \int d^3\vec{k} \sum_{\sigma} (N_{k\sigma}+1) \left| \langle \psi_f | \vec{A}^{*}_{k\sigma}(\vec{r}) \left(\frac{-q\hat{\vec{p}}}{m}\right) | \psi_i \rangle \right|^2$$

$$\cdot \, \delta(E_f - E_i + \hbar\omega_k)$$

$$= \frac{2\pi}{\hbar} \frac{V}{(2\pi)^3} \frac{\hbar}{2\varepsilon_0 V} \sum_{\sigma} \int d^3\vec{k} \, \frac{N_{k\sigma}+1}{\omega_k} \left| \langle \psi_f | \hat{e}^{*}_{k\sigma} e^{-i\vec{k}\cdot\vec{r}} \left(\frac{q}{m}\hat{\vec{p}}\right) | \psi_i \rangle \right|^2$$

$$\cdot \, \delta(E_f - E_i + \hbar\omega_k).$$

The matrix element

$$M_{fi} = \langle \psi_f | \frac{-q\hat{\vec{p}}}{m} e^{-i\vec{k}\cdot\vec{r}} | \psi_i \rangle \cdot \vec{e}^{\,*}_{k\sigma} =$$

$$= \int d^3\vec{r} \left(\psi_f^*(\vec{r}) \frac{i\hbar q}{m} \vec{\nabla} \psi_i(\vec{r}) e^{-i\vec{k}\cdot\vec{r}} \right) \cdot \vec{e}^{\,*}_{k\sigma}$$

$\psi_{i,f}(\vec{r})$ is localized on the size of atom, r_{atom}, which is much smaller that wavelength λ of the photon,

$$\Rightarrow \vec{r}\cdot\vec{k} \sim \frac{r_{atom}}{\lambda} \ll 1 \Rightarrow e^{-i\vec{k}\cdot\vec{r}} \approx 1 \quad \left(\text{dipole approximation!}\right)$$

Next observe that $\frac{-i\hbar\vec{\nabla}}{m} = \frac{\hat{\vec{p}}}{m}$ = velocity operator $= \hat{V}$

But $\hat{V} = \dot{\vec{r}} = \frac{i}{\hbar}[\hat{H}, \vec{r}]$ (equation of motion)

$$\Rightarrow \langle \psi_f | -\frac{i\hbar}{m}\vec{\nabla} | \psi_i \rangle = \frac{i}{\hbar} \langle \psi_f | [\hat{H}, \vec{r}] | \psi_i \rangle =$$

$$= \frac{i}{\hbar} \langle \psi_f | \hat{H}_0 \vec{r} - \vec{r}\hat{H}_0 | \psi_i \rangle =$$

$$= \frac{i}{\hbar}(E_f - E_i) \langle \psi_f | \vec{r} | \psi_i \rangle$$

$\Rightarrow$ the matrix element of current $\frac{q}{m}\hat{p}$ reduces to that of coordinate.

Moreover $\langle \psi_f | q\vec{r} | \psi_i \rangle = \int d^3\vec{r}\, q\, \psi_f^*(\vec{r})\, \vec{r}\, \psi_i(\vec{r}) =$

$= \vec{d}_{fi}$ = electric dipole matrix element.

$$\Rightarrow M_{fi} = \frac{i}{\hbar} \underbrace{(E_f - E_i)}_{-\hbar\omega_k \text{ due to } \delta(E_f - E_i + \hbar\omega_k)} \vec{d}_{fi} \cdot \vec{e}^{\,*}_{k\sigma} = -i\,\omega_k\, \vec{d}_{fi} \cdot \vec{e}^{\,*}_{k\sigma}$$

$$W_{fi}^{emi} = \frac{1}{2\varepsilon_0 (2\pi)^2} \sum_\sigma \int d^3\vec{k}\, (N_{k\sigma}+1)\, \omega_k \left|\vec{d}_{fi}\cdot \vec{e}^{\,*}_{k\sigma}\right|^2 \delta(E_f - E_i + \hbar\omega_k)$$

$$d^3\vec{k} = k^2 dk\, d\Omega_k = \frac{1}{c^3}\, \omega_k^2 d\omega_k d\Omega_k \quad \text{since } \omega_k = ck$$

$$W_{fi}^{emi} = \frac{1}{8\pi^2\varepsilon_0 c^3} \int d\Omega \int_0^\infty d\omega\, \omega^3 (N_\omega + 1) \sum_\sigma \left|\vec{d}_{fi}\cdot \vec{e}^{\,*}_{k\sigma}\right|^2 \cdot$$
$$\cdot\, \delta\left(\hbar\left(\frac{E_f - E_i}{\hbar} + \omega\right)\right)$$

$$\left(\omega_{if} = \frac{E_i - E_f}{\hbar}\right)$$

$$= \frac{\omega_{fi}^3}{8\pi^2\varepsilon_0 c^3 \hbar} \int d\Omega_k\, (N_\omega + 1) \sum_\sigma \left|\vec{d}_{fi}\cdot \vec{e}^{\,*}_{k\sigma}\right|^2$$

here $N_\omega = N_{k=\omega/c}$ and we assumed that occupation number $N_{\vec{k},\sigma}$ is independent of polarization σ.

With the help of "polarization note" we can write

$$\sum_\sigma \left|\vec{d}\cdot \vec{e}^{\,*}_{k\sigma}\right|^2 = \sum_\sigma \sum_a d_a (e^*_{k\sigma})_a \sum_b d_b^* (e_{k\sigma})_b =$$

$$= \left(\text{but } \sum_\sigma (e^*_{k\sigma})_a (e_{k\sigma})_b = \delta_{ab} - \hat{k}_a \hat{k}_b\right) =$$

$$= \sum_{ab} d_a d_b^* (\delta_{ab} - \hat{k}_a \hat{k}_b) = \vec{d}\cdot\vec{d}^{\,*} - (\vec{d}\cdot\hat{k})(\vec{d}^{\,*}\cdot\hat{k})$$

$$= \left|\vec{d}\times\hat{k}\right|^2 \quad \text{where } \hat{k} = \hat{n} = \text{unit vector along } \vec{k}.$$

$$W_{fi}^{emi} = \frac{(N_\omega + 1)\,\omega_{fi}^3}{8\pi^2 \varepsilon_0 c^3 \hbar} \underbrace{\int d\Omega_k \, |\vec{d} \times \hat{n}|^2}_{\underbrace{\int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \cdot \sin^2\theta}_{\frac{8\pi}{3}} \, |d_{fi}|^2}$$

$$W_{fi}^{emi}(\omega) = \frac{(N_\omega + 1)\,\omega^3}{3\pi \varepsilon_0 c^3 \hbar} \, |\vec{d}_{fi}|^2$$

where $\vec{d}_{fi} = \langle \psi_f | q \vec{r} | \psi_i \rangle$ is electric dipole matrix element.

Observe now that emitted energy is given by

$$W_{fi}^{emi} \cdot \hbar\omega = (N_\omega + 1) \, \frac{\omega^4 \, |\vec{d}_{fi}|^2}{3\pi \varepsilon_0 c^3}$$

Stimulated emission Spontaneous emission

Consideration of <u>absorption</u> is very similar

$$W_{fi}^{abs} = \frac{2\pi}{\hbar} \sum_{\vec{k}\sigma} |(\hat{U}_{k\sigma})_{fi}|^2 \, \delta(E_f - E_i - \hbar\omega_k)$$

and $|i\rangle = \psi_i \, \phi_N$

$|f\rangle = \psi_f \, \phi_{N-1}$